CBCS SCHEME

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C.M.R. INSTITUTE OF TECHNOLOGY CMR BANGALORE

First Semester B.E./B.Tech. Degree Examination, Jan./Feb. 2023
Applied Physics for CSE Stream

Time: hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.

		Module – 1	M	L	C
Q.1	a.	Explain the construction and working of a semiconductor laser with the help of energy level diagram.	8	L2	CO1
	b.	What is refractive index profile? Discuss three different types of optical fibres based on modes of propagation and refractive index profile.	7	L2	CO1
	c.	The angle of acceptance of an optical fibre is 30° when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33.	5	L3	CO5
		OR			
Q.2	a.	Define acceptance angle and numerical aperture. Derive an expression for numerical aperture in terms of refractive indices of core, cladding and surrounding.	8	L2	CO1
	b.	Derive an expression for energy density for a system in thermal equilibrium in terms of Einstein's co-efficient.	7	L2	CO1
	c.	In a diffraction grating experiment the laser light undergoes second order diffraction for diffraction angle 1.48°. The grating constant is 5.08×10^{-5} m and the distance between the grating and the source is 80 cm, find the wave length of LASER light.	5	L3	CO5
		Module – 2			
Q.3	a.	Assuming the time independent Schrodinger's wave equation discuss the solution for a particle in one dimensional potential well of infinite height and hence obtain the normalized wave equation.	9	L2	CO2
	b.	State and explain Heisenberg uncertainty principle. Show that an electron doesn't exists inside the nucleus.	7	L2	CO2
	c.	Compute the deBroglie wavelength for a neutron moving with one tenth part of velocity of light. Given the mass of the neutron is 1.674×10^{-27} kg.	4	L3	CO2
		OR			
Q.4	a.	Setup Schrodinger time independent wave equation in one dimension.	9	L2	CO2
	b.	Define phase velocity and group velocity. Derive an expression for De Broglie wavelength of an electron.	7	L2	CO2
	c.	An electron has a speed of 100 m/s. The inherent uncertainity in its measurement is 0.005%. Calculate the corresponding uncertainity in the measurement of the position.	4	L3	CO2
		Module – 3			
Q.5	a.	Explain the Pauli matrices and apply Pauli matrices on the state $\|0\rangle$ and $\|1\rangle$.	9	L2	CO2
	b.	Differentiate between classical and quantum computing.	6	L2	CO2
	c.	Explain the Hadamard gate. Show that the Hadamard gate is unitary.	5	L2	CO2

		OR			
Q.6	**a.**	A Linear operator 'X' operates such that $X\|0\rangle = \|1\rangle$ and $X\|1\rangle = \|0\rangle$. Find the matrix representation of 'X'.	5	L2	CO2
	b.	Describe the working of CNOT gate mentioning its matrix representation and truth table.	9	L2	CO2
	c.	Explain the representation of qubit using Bloch sphere.	6	L2	CO2
		Module – 4			
Q.7	**a.**	Enumerate the failures of classical free electron theory and discuss the success of quantum free electron theory of metals.	8	L2	CO3
	b.	Explain DC and AC Josephson effects and mention any two applications of superconductivity in quantum computing.	7	L2	CO3
	c.	Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above the Fermi energy is occupied.	5	L3	CO3
		OR			
Q.8	**a.**	Explain Meissner's effect and the variation of critical field with temperature.	8	L2	CO3
	b.	Define Fermi factor. Discuss the variation of Fermi factor with temperature and energy.	7	L2	CO3
	c.	The critical temperature of Nb is 9.15 K. At zero Kelvin, the critical field is 0.196T. Calculate the critical field at 8 K.	5	L3	CO3
		Module – 5			
Q.9	**a.**	Discuss timing in Linear motion, Uniform motion, Slow in and Slow out.	8	L2	CO4
	b.	Describe Jumping and parts of jumping in animation.	7	L2	CO4
	c.	A slowing-in object in an animation has a first frame distance 0.5 m and first slow in frame 0.35 m. Calculate the base distance and the number of frames in sequence.	5	L3	CO5
		OR			
Q.10	**a.**	Illustrate the odd rule and odd rule multiplier with suitable example.	8	L2	CO4
	b.	Discuss modeling the probability for proton decay.	7	L2	CO4
	c.	In an optical fibre experiment the Laser light propagating through optical fibre cable of 1.5 m, made a spot diameter of 8 mm on the screen. The distance between the end of the optical fibre cable and the screen is 3.4 cm. Calculate the angle of contact and numerical aperture of given optical fibre.	5	L3	CO5

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CBCS SCHEME

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BPH

4/18

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2023

Physics for CSE Stream

Time: 3 hrs. Max. Marks: 100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.
4. Constants : Speed of Light $C = 3 \times 10^8$ m/s, Boltzmam const. $K = 1.38 \times 10^{-23}$ J/K, Planck's const $h = 6.625 \times 10^{-34}$ JS, Acceleration due to gravity $g = 9.8 m/s^2$, mass of electron $m = 9.1 \times 10^{-31}$ Kg

		Module – 1	M	L	C
Q.1	a.	Derive an expression for energy density interms of Einstein's coefficients in Laser action.	10	L2	CO1
	b.	Explain types of optical fibers.	6	L2	CO1
	c.	The ratio of population inversion of two energy levels is 1.059×10^{-30}. Find the wavelength of Light emitted by spontaneous emissions at 330K.	4	L3	CO1
		OR			
Q.2	a.	Derive an expression for Numerical aperture in an optical fiber.	8	L2	CO1
	b.	Discuss construction and working of semiconductor diode Laser with energy level diagram.	8	L2	CO1
	c.	The angle of acceptance of an optical fiber is 30°, when kept in air. Find the angle of acceptance when it is in a medium of refractive index 1.33.	4	L3	CO1
		Module – 2			
Q.3	a.	What is wave packet? Give physical significance and properties of wave function? Define group velocity.	8	L2	CO1
	b.	State and explain Heisenberg's uncertainty principle. Give its physical significance. Show that electron cannot exist inside the nucleus.	8	L2	CO2
	c.	A particle of mass 0.5meV/c^2 has kinetic energy 100eV. Find its de Broglie wavelength, where 'C' is the velocity of light.	4	L3	CO2
		OR			
Q.4	a.	Derive an expression for Schrödinger's Time independent equation one dimensional form.	8	L2	CO2
	b.	Obtain the expression for energy eigen values using Schrodinger's time independent equation.	8	L2	CO2
	c.	In a measurement of position and velocity of an electron moving with a speed of 6×10^5 m/s, calculate highest accuracy with which its position could be determined, if the inherent error in the measurement of its velocity is 0.01% for the speed stated.	4	L3	CO2

		Module – 3			
Q.5	a.	Explain single qubit gate and multiple qubit gate with example for each.	8	L2	CO2
	b.	Discuss CNOT gate and its operation on four different input states.	8	L2	CO2
	c.	Given $A = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$ prove that $A^* = A$.	4	L3	CO2
		OR			
Q.6	a.	Elucidate the differences between classical computing and Quantum computing.	8	L2	CO2
	b.	Discuss the working of phase gate mentioning its matrix representation and truth table.	8	L2	CO2
	c.	Find the inner product of states $\|1\rangle$ and $\|0\rangle$ and draw conclusionon the result.	4	L3	CO2
		Module – 4			
Q.7	a.	Distinguish between Type – I and Type – II super conductors.	8	L2	CO3
	b.	Discuss the effect of temperature and impurity on electrical resistivity of conductors and hence explain for superconductors.	8	L2	CO3
	c.	In a diffraction grating experiment the laser light undergoes second order diffraction, if the distance between screen and grating is 20cm, and average distance of 2nd order spot 2.7cm grating constant 1×10^{-3}m, calculate the wavelength of laser light.	4	L3	CO5
		OR			
Q.8	a.	Explain B.C.S theory of superconductivity.	7	L2	CO1
	b.	Define Fermi energy level. Discuss various energy states by the electrons at T = O K and T > O K on the basis of fermifactor.	8	L2	CO1
	c.	Calculate the acceptance angle and numerical aperture of given optical fiber having diameter of spot is 2.6cm and distance between screen and optical fiber 3.0cm.	5	L2	CO1
		Module – 5			
Q.9	a.	Elucidate the importance of size and scale and weight and strength in animations.	8	L2	CO4
	b.	Discuss modeling probability of proton decay.	8	L2	CO4
	c.	The number of particles emitted per second by a random radioactive source has a Poisson's distribution with $\lambda = 4$. Calculate the probability of P(X = 0) and P(X = 1)	4	L3	CO4
		OR			
Q.10	a.	Discuss timing in Linear motion, uniform motion, show in and flow out.	8	L2	CO4
	b.	Discuss salient features of Normal distribution using Bell curves.	8	L2	CO4
	c.	A slowing in object in an animation has a first frame distance 0.5m and first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.	4	L3	CO4

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MAKE-UP EXAM

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BPHY

6/18

First/Second Semester B.E./B.Tech. Degree Examination, Nov./Dec. 2023

Applied Physics for CSE Stream

Time: 3 hrs. Max. Marks: 100

Note: *1. Answer any FIVE full questions, choosing ONE full question from each module.*
2. VTU Formula Hand Book is permitted.
3. M : Marks , L: Bloom's level , C: Course outcomes.
4. Physical constants: Plank's constant, $h = 6.625 \times 10^{-34}$ J-S; Speed of light, $c = 3 \times 10^{8}$ ms^{-1}; Mass of electron, $m = 9.1 \times 10^{-31}$ kg; Charge of electron, $e = 1.6 \times 10^{-19}$ C; Boltzmann constant, $k = 1.38 \times 10^{-23}$ JK^{-1}; Acceleration due to gravity, $g = 9.8$ ms^{-2}.

		Module – 1	M	L	C
Q.1	a.	Obtain the expression for energy density of radiation under thermal equilibrium condition interms of Einstein's co-efficient.	09	L2	CO1
	b.	Explain the different types of optical fibers with suitable diagrams.	06	L2	CO1
	c.	An optical fiber of length 2 km has input power of 200 mW, which emerges out with power of 160 mW. Calculate the attenuation co-efficient of the fiber.	05	L3	CO1
		OR			
Q.2	a.	Describe the construction, principle and working of a semiconductor LASER with neat diagrams.	07	L2	CO1
	b.	Define numerical aperture and acceptance angle. Obtain an expression for numerical aperture interms of refractive indices of core, cladding and surrounding medium.	08	L2	CO1
	c.	In diffraction grating experiment the LASER light undergoes first order diffraction with diffracting angle 23.86°. The grating constant is 1.66×10^{-6} m^{-1}. Calculate the wavelength of LASER source.	05	L3	CO5
		Module – 2			
Q.3	a.	State Heisenberg's uncertainty principle. Using this principle, prove that the electron does not exists inside the nucleus.	08	L2	CO2
	b.	Set up one dimensional time-independent Schrodinger wave equation.	08	L2	CO2
	c.	An electron is bound in one dimensional infinite potential well of width 0.12 nm. Find the energy value and de-Broglie wavelength in first excited level.	04	L3	CO2
		OR			
Q.4	a.	State and explain de-Broglie's hypothesis and derive the expression for de-Broglie wavelength by analogy.	06	L2	CO2
	b.	Derive the expression for energy eigen functions and eigen values for a particle in one dimensional infinite potential well.	10	L2	CO2
	c.	Estimate the potential difference through which an electron is needed to be accelerated so that its de-Broglie wavelength becomes equal to 20 Å.	04	L3	CO2
		Module – 3			
Q.5	a.	Define single and two qubits. Explain the block sphere representation of qubit.	08	L2	CO2
	b.	Explain the controlled NOT gate (CNOT gate) with four different input states with the truth table.	08	L2	CO2
	c.	Show that S - gate can be formed by connecting two T – gates in series.	04	L3	CO2
		OR			

Q.6	a.	Mention the Pauli's matrices. Discuss the operations of Pauli's matrices on | 0 > and | 1 > states.	10	L2	CO2
	b.	Explain the operations of phase gate [S – gate] with | 0 > and | 1 > states with truth table. Mention its matrix representation.	06	L2	CO2
	c.	A linear operator X operates such that X | 0 > = | 1 > and X |1 > = | 0 >. Find the matrix representation of the operator X.	04	L3	CO2
		Module – 4			
Q.7	a.	What is Meissner effect? Explain Type I [Soft] and Type II [Hard] super conductors.	08	L2	CO3
	b.	What is Fermi factor? Discuss the variation of Fermi factor with temperature and energy.	08	L2	CO3
	c.	Superconducting tin has a critical magnetic field of 0.0217 T at 2K. If the critical temperature for superconducting transition for tin is 3.7K, find the critical magnetic field at 3K.	04	L3	CO3
		OR			
Q.8	a.	Define critical temperature and critical magnetic field. Explain briefly BCS theory of superconductivity.	08	L2	CO3
	b.	Enumerate the failures of classical free electron theory and mention the assumptions of quantum free electron theory.	08	L2	CO3
	c.	The Fermi level in potassium is 2.1 eV. What is the energy of the energy level for which the probability of occupation at 300 K is 0.98?	04	L3	CO3
		Module – 5			
Q.9	a.	Explain the odd rule with odd rule multipliers with suitable examples.	08	L2	CO4
	b.	Explain Poisson distribution and probability mass function with example.	07	L2	CO4
	c.	In case of animating a jump, the push height is 0.5m and jump magnification is 5. Calculate the jump height and push acceleration.	05	L3	CO4
		OR			
Q.10	a.	Discuss slow in and slow out with neat diagrams.	08	L2	CO4
	b.	Write a note on Monte-Carlo method and discuss the determination of the value of π using Monte-Carlo method.	07	L2	CO4
	c.	In an optical fiber experiment the LASER light propagating through optical made a spot diameter of 21 mm on the screen. When the distance between the end of the fiber and the screen is 31 mm, calculate the acceptance angle and numerical aperture.	05	L3	CO5

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CBCS SCHEME

USN | | | | | | | | | | BPHYS102/

8/18

First/Second Semester B.E./B.Tech. Degree Examination, Dec.2023/Jan.2024
Applied Physics for CSE Stream

Time: 3 hrs. Max. Marks: 100

Note: 1. *Answer any FIVE full questions, choosing ONE full question from each module.*
2. *Draw neat sketches where ever necessary.*
3. *VTU Formula Hand Book is permitted.*
4. *M : Marks , L: Bloom's level , C: Course outcomes.*
5. *Constants: Speed of Light C = 3 × 10^8 m/s, Boltzmam const. K = 1.38×10^{-23} J/K^{-1}, Planck's const h = 6.625 × 10^{-34} JS, Acceleration due to gravity g = 9.8m/s^{-2}, Permittivity of Free space ϵ_0 = 8.854×10^{-12} Fm^{-1}*

		Module – 1	M	L	C
Q.1	a.	Define LASER and explain the interaction of radiation with matter for the induced absorption, spontaneous emission and stimulated emission.	7	L1	CO1
	b.	Discuss different types optical fibers based on modes of propagation and refractive index profile.	9	L2	CO1
	c.	Find attenuation in an optical fiber of length 500m, when a light signal of power 100mW emerges out of the fiber with a power of 90mW.	4	L3	CO1
		OR			
Q.2	a.	Obtain the expression for energy density of radiation using Einstein's co-efficient A and B and thus conclude $B_{12} = B_{21}$.	9	L2	CO1
	b.	Discuss point to point communication using optical fiber.	6	L2	CO1
	c.	In a diffraction grating experiment the LASER light undergoes second order diffraction for diffraction angle 1.48°. The grating constant d = 5.05 × 10^{-3} m and the distance between the grating and source is 0.60m, find the wavelength of LASER light.	5	L3	CO5
		Module – 2			
Q.3	a.	State and explain Heisenberg's uncertainty principle. Using the principle show that electron doesn't exist inside the nucleus.	7	L2	CO2
	b.	Set up Schrodinger's time independent wave equation in one dimension.	8	L2	CO2
	c.	A particle of mass 0.5 MeV/C^2 has kinetic energy 100 eV. Find its de-Broglie wavelength where 'C' is the velocity of light.	5	L3	CO2
		OR			
Q.4	a.	Find the Eigen values and Eigen functions for a particle in one dimensional infinite potential well.	9	L2	CO2
	b.	Discuss de-Broglie hypothesis.	6	L2	CO2
	c.	Calculate the energy of the first three states for an electron in one dimensional potential well of width 1A°.	5	L3	CO2

		Module – 3			
Q.5	a.	Explain the representation of qubit using Bloch sphere.	6	L2	
	b.	Discuss CNOT gate, matrix representation and its operation on four different input states.	6	L2	
	c.	A linear operator X operates such that $X\|0\rangle = \|1\rangle$ and $X\|1\rangle = \|0\rangle$. Find the matrix representation of X	8	L3	CO2
		OR			
Q.6	a.	State the Pauli's metrics and apply Pauli matrices on the states $\|0\rangle$ and $\|1\rangle$ states.	8	L3	CO2
	b.	Elucidate the differences between classical and quantum computing.	6	L2	CO2
	c.	Explain matrix representation of 0 and 1 states and apply identify operator I to $\|0\rangle$ and $\|1\rangle$ states.	6	L3	CO2
		Module – 4			
Q.7	a.	Enumerate the failures of Classical Free Electron [CFET] Theory and mention the assumptions of Quantum Free Electron Theory [QFET]	7	L2	CO3
	b.	Describe Meissner's effect. Distinguish between Type I and Type II super conductors.	8	L2	CO3
	c.	Lead has a superconducting transition temperature of 7.26K. If initial field at 0K is $50 \times 10^3 Am^{-1}$, calculate the critical field at 6K.	5	L3	CO3
		OR			
Q.8	a.	Define Fermi factor. Discuss the variation of Fermi factor with temperature and energy.	7	L2	CO3
	b.	Explain the phenomenon of super conductivity. Discuss qualitatively BCS theory of super conductivity.	8	L2	CO3
	c.	Calculate the probability of occupation of an energy level 0.02eV above Fermi level at temperature 200K.	5	L3	CO3
		Module – 5			
Q.9	a.	Discuss timing in linear motion, uniform motion, slow in and slow out.	8	L2	CO4
	b.	Enumerate the difference between inferential and descriptive statistics.	6	L2	CO4
	c.	In an optical fiber experiment, the light passing through the fiber, made a spot diameter of 8mm on the screen. The distance between the end of the optical fiber cable and the screen is 0.031m. Calculate the angle of acceptance and numerical aperture of given optical fiber.	6	L3	CO5
		OR			
Q.10	a.	Describe Jumping and parts of Jump.	8	L2	CO4
	b.	Discuss the salient features of Normal distribution using Bell curves.	7	L2	CO4
	c.	While animating speeding up car animation, the total distance covered over 6 frames is 25m. Calculate the base distance.	5	L3	CO4

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22PHYS12/22

Model Question Paper-I with effect from 2022-23 (CBCS Sche

10/18

USN | | | | | | | | | | |

First/Second Semester B.E. Degree Examination

Applied Physics for Computer Science Stream

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE**.
02. Draw neat sketches where ever necessary.
03. **Constants :** Speed of Light 'c' = 3 $\times 10^{8}$ ms^{-1}, Boltzmann Constant 'k' = 1.38 $\times 10^{-23}$ JK^{-1}, Planck's Constant 'h' = 6.625 $\times 10^{-34}$ Js, Acceleration due to gravity 'g'= 9.8 ms^{-2}, Permittivity of free space 'ε_0'=8.854 $\times 10^{-12}$ F m^{-1}.

		Module -1	*Bloom's Taxonomy Level	Marks
Q.01	a	Define LASER and Discuss the interaction of radiation with matter.	L2	7
	b	Define Acceptance angle and Numerical Aperture and hence derive an expression for NA in terms of RIs core, cladding and surrounding.	L2	8
	c	A LASER source has a power output of 10^{-3} W. Calculate the number of photons emitted per second given the wavelength of LASER 692.8 nanometer.	L3	5
		OR		
Q.02	a	Illustrate the construction and working of Semiconductor LASER with a neat sketch and energy level diagram also mention its applications.	L2	9
	b	Discuss the types of optical fibers based on Modes of Propagation and RI profile.	L2	6
	c	Obtain the attenuation co-efficient of the given fiber of length 1500 m given the input and output power 100 mW and 70 mW.	L3	5
		Module-2		
Q.03	a	Setup SchrÖdinger time independent wave equation in one dimension.	L2	8
	b	State and Explain Heisenberg's Uncertainty principle and Principle of Complementarity.	L2	7
	c	An electron is kinetic energy 500 keV is in vacuum. Calculate the group velocity and de Broglie wavelength assuming the mass of the moving electron is equal to the rest mass of electron.	L3	5
		OR		
Q.04	a	Discuss the motion of a quantum particle in a one-dimensional infinite potential well of width 'a' and also obtain the eigen functions and energy eigen states.	L2	10
	b	Explain the physical significance of the Wave Function.	L2	5
	c	The speed of electron is measured to with in an uncertainty of 2 $\times$ 10^{4} ms^{-1} in one dimension. What is the minimum width required by the electron to be confined in an atom?	L3	5
		Module-3		
Q.05	a	Define a bit and qbit and explain the properties of qubit.	L2	6
	b	Discuss the CNOT gate and its operation on four different input states.	L2	6
	c	A Linear Operator 'X' operates such that X $\|0\rangle$ = $\|1\rangle$ and X $\|1\rangle$ = $\|0\rangle$. Find the matrix representation of 'X'.	L3	8
		OR		
Q.06	a	State the Pauli matrices and apply Pauli matrices on the states $\|0\rangle$ and $\|1\rangle$.	L2	8
	b	Elucidate the differences between classical and quantum computing.	L2	6

c	Describe the working of controlled-Z gate mentioning its matrix representation and truth-table.	L3
	Module-4	
a	Define Fermi Factor and Discuss the variation of Fermi factor with temperature and energy.	L2
b	Explain DC and AC Josephson effects and mention the applications of superconductivity in quantum computing.	L2
c	Calculate the probability of occupation of an energy level 0.2 eV above fermi level at temperature 27°C.	L3
	OR	
a	Describe Meissner's Effect and hence classify superconductors into Soft and Hard superconductors using M-H graphs.	L2
b	Enumerate the assumptions of Quantum free Electron Theory of Metals	L2
c	Lead has superconducting transition temperature of 7.26 K. If the initial field at 0K is 50×10^3 Am^{-1} Calculate the critical field at 6k.	L3
	Module-5	
a	Discuss timing in Linear motion, Uniform motion, slow in and slow out.	L2
b	Distinguish between descriptive and inferential statistics.	L2
c	Illustrate the odd rule and odd rule multipliers with a suitable example.	L3
	OR	
a	Describe Jumping and parts of jump.	L2
b	Discuss the salient features of Normal distribution using bell curves.	L2
c	The number of particles emitted per second by a random radioactive source has a Poisson's distribution with $\lambda = 4$. Calculate the probability of $P(X = 0)$ and $P(X = 1)$.	L3

n's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs
l by every bit of questions.

Model Question Paper-II with effect from 2022-23 (CBCS Scheme)

USN | | | | | | | | | | |

12/18

First/Second Semester B.E. Degree Examination

Applied Physics for Computer Science Stream

TIME: 03 Hours **Max. Marks: 100**

Note: 01. Answer any **FIVE** full questions, choosing at least **ONE** question from each **MODULE.**
02. Draw neat sketches where ever necessary.
03. **Constants :** Speed of Light 'c' = 3 $\times 10^{8}$ ms^{-1}, Boltzmann Constant 'k' = 1.38 $\times 10^{-23}$ JK^{-1}, Planck's Constant 'h' = 6.625 $\times 10^{-34}$ Js, Acceleration due to gravity 'g'= 9.8 ms^{-2}, Permittivity of free space 'ε_0'=8.854 $\times 10^{-12}$ F m^{-1}.

		Module -1	*Bloom's Taxonomy Level	Marks
Q.01	a	Obtain the expression for Energy Density using Einstein's A and B coefficients and thus conclude on $B_{12}=B_{21}$.	L2	8
	b	Describe attenuation and explain the various fiber losses.	L2	7
	c	Given the Numerical Aperture 0.30 and RI of core 1.49 Calculate the critical angle for the core-cladding interface.	L3	5
		OR		
Q.02	a	Discuss the applications of LASER in bar-code scanner and LASER Cooling.	L2	9
	b	Discuss Point to Point communication using optical fibers.	L2	6
	c	Calculate the ratio of population for a given pair of energy levels corresponding to emission of radiation 694.3 nm at a temperature of 300 K.	L3	5
		Module-2		
Q.03	a	Derive an expression for de Broglie wavelength by analogy and hence discuss the significance of de Broglie waves.	L2	6
	b	Explain the Wave function with mathematical form and Discuss the physical significance of a wave function.	L2	9
	c	Calculate the energy of the first three states for an electron in one dimensional potential well of width 0.1 nm.	L3	5
		OR		
Q.04	a	Explain Eigen functions and Eigen Values and hence derive the eigen function of a particle inside infinite potential well of width 'a' using the method of normalization.	L2	10
	b	Show that electron does not exist inside the nucleus using Heisenberg's uncertainty principle.	L2	5
	c	An electron is associated with a de Broglie wavelength of 1nm. Calculate the energy and the corresponding momentum of the electron.	L3	5
		Module-3		
Q.05	a	Discuss the working of phase gate mentioning its matrix representation and truth table.	L2	6
	b	Explain Orthogonality and Orthonormality with an example for each.	L2	6
	c	Given $\|\phi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$ and $\|\phi\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$ Prove that $\langle\psi\|\phi\rangle = \langle\phi\|\psi\rangle^*$	L3	8
		OR		
Q.06	a	Explain the representation of qubit using Bloch Sphere.	L2	6
	b	Explain Single qubit gate and multiple qubit gate with an example for each.	L2	8

	c	Explain the Matrix representation of 0 and 1 States and apply identity operator I to | 0 ⟩ and | 1 ⟩ states.		6
		Module-4		
Q.07	a	Enumerate the failures of classical free electro theory and assumptions of quantum free electron theory of metals.	L2	7
	b	Explain Meissner's Effect and the variation of critical field with temperature.	L2	8
	c	A superconducting tin has a critical temperature of 3.7 K at zero magnetic field and a critical field of 0.0306 Tesla at 0 K. Find the critical field at 2 K.	L3	5
		OR		
Q.08	a	Explain the phenomenon of superconductivity and Discuss qualitatively the BCS theory of superconductivity for negligible resistance of metal at temperatures close to absolute zero.	L2	9
	b	Give the qualitative explanation of RF Squid with the help of a neat sketch.	L2	6
	c	Find the temperature at which there is 1% probability that a state with an energy 0.5 eV above Fermi energy is occupied.	L3	5
		Module-5		
Q.09	a	Elucidate the importance of size & scale and weight and strength in animations.	L2	8
	b	Mention the general pattern of monte Carlo method and hence determine the value of π.	L2	6
	c	Describe the calculation of Push time and stop time with examples.	L3	6
		OR		
Q.10	a	Sketch and explain the motion graphs for linear, easy ease,easy ease in and easy ease out cases of animation.	L2	8
	b	Discuss modeling the probability for proton decay.	L2	7
	c	A slowing-in object in an animation has a first frame distance 0.5m and the first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.	L3	5

*Bloom's Taxonomy Level: Indicate as L1, L2, L3, L4, etc. It is also desirable to indicate the COs and POs to be attained by every bit of questions.

CBCS SCHEME

USN | | | | | | | | | | |

BPHYS 14/18

First / Second Semester B.E./B.Tech. Degree Examination, Dec.2024/Jan.2025

Applied Physics for CSE Stream

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. VTU Handbook is permitted.

		Module – 1	M	L	C
Q.1	a.	Obtain the expression for energy density equation using Einstein's coefficients at thermal equilibrium condition.	9	L2	CO1
	b.	Discuss the types of optical fibers based on modes of Propagation and Refractive Index profile.	6	L2	CO1
	c.	Given the numerical aperture 0.30 and RI f core 1.49. Calculate the critical angle for the core – cladding interface.	5	L3	CO1
		OR			
Q.2	a.	Illustrate the construction and working of semiconductor laser with a neat sketch with energy level diagram.	8	L2	CO1
	b.	Define Acceptance angle and Numerical aperture and hence derive an expression for Numerical aperture in terms of Refractive Indices of core, cladding and surrounding.	7	L2	CO1
	c.	In an optical fiber experiment the LASER light propagating through optical fibre cable of 1.5m , made a spot diameter of 8mm on the screen. The distance between the end of the optical fibre cable and the screen is 0.031m. Calculate angle of contact and N.A of given optical fibre.	5	L3	CO5
		Module – 2			
Q.3	a.	Derive an expression for dc Broglie wavelength by analogy and hence discuss the significance of de Broglie waves.	6	L2	CO2
	b.	Set – up Schrodinger time independent wave equation in one dimension.	9	L2	CO2
	c.	Calculate the energy of the first three energy states for an electron in one dimensional potential well of width 0.1nm.	5	L3	CO2
		OR			
Q.4	a.	State and explain Heisenberg's uncertainty principle and show that electron does not exist inside the nuclear using Heisenberg's uncertainty principle.	7	L2	CO2
	b.	Explain Eigen values and Eigen functions and hence derive the Eigen function of a particle inside infinite potential well of width 'a' using the method of Normalization.	8	L2	CO2

	c.	The kinetic energy of an electron is equal to the energy of a photon with a wave length of 560 nm. Calculate the de Broglie wave length of the electron.	5	L3	
		Module – 3			
Q.5	a.	State the Pauli matrices and apply Pauli matrices on the states $\|0\rangle$ and $\|1\rangle$.	9	L2	CO2
	b.	Discuss the CNOT gate and its operation on four different input states.	6	L2	CO2
	c.	Given $\|\Psi\rangle = \begin{pmatrix} \alpha_1 \\ \alpha_2 \end{pmatrix}$ and $\|\phi\rangle = \begin{pmatrix} \beta_1 \\ \beta_2 \end{pmatrix}$, Prove that $\langle\Psi\|\phi\rangle = \langle\phi\|\Psi\rangle^*$.	5	L3	CO2
		OR			
Q.6	a.	Explain the representation of qubit using Block sphere.	7	L2	CO3
	b.	Describe the working of controlled – Z gate mentioning its matrix representation and truth table.	8	L3	CO3
	c.	A linear operator 'X' operates such that $X\|0\rangle = \|1\rangle$ and $X\|1\rangle = \|0\rangle$. Find the matrix representation of 'X'.	5	L3	CO3
		Module – 4			
Q.7	a.	Enumerate the assumptions of Quantum free electron theory of metals and mention the failures and classical free electron theory.	7	L2	CO3
	b.	Describe Meissner effect and hence classify superconductors into Soft and Hard super conductors using M – H graphs.	8	L2	CO3
	c.	Calculate the probability of occupation of an energy level 0.2ev above Fermi level at temperature 27°C.	5	L3	CO3
		OR			
Q.8	a.	Define Fermi factor and discuss the variation of Fermi factor with temperature and energy.	7	L2	CO3
	b.	Explain the phenomenon of superconductivity and discuss qualitatively the BCS theory of super conductivity for negligible resistance of metal at temperature close to absolute zero.	8	L2	CO3
	c.	A superconductivity Tin has a critical temperature of 3.7K at zero magnetic field and a critical field of 0.0306 Tesla at 0°K. Find the critical field at 2K.	5	L3	CO3
		Module – 5			
Q.9	a.	Elucidate the importance of size and scale , weight and strength in animations.	8	L2	CO4
	b.	Discuss the salient features of normal distribution using bell curves.	7	L2	CO4

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	c.	A slowing object in an animations has a first frame distance 0.5m and the first slow in frame 0.35m. Calculate the base distance and the number of frames in sequence.	5	L3	CO4
		OR			
Q.10	a.	Describe Jumping and parts of Jump.	8	L2	CO4
	b.	Discuss modeling the probability for proton decay.	7	L2	CO4
	c.	In a diffraction grating experiment the laser light undergoes second order diffraction for diffraction angle 1.48°. The grating constant $d = 5.05 \times 10^{-5}$m and the distance between the grating and screen is 0.60m. Find the wavelength of LASER light.	5	L3	CO5

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CBCS SCHEME

17/18

USN

BPHYS102/202

First/Second Semester B.E./B.Tech. Degree Examination, June/July 2024
Applied Physics for CSE Stream

Time: 3 hrs. Max. Marks: 100

Note: 1. Answer any FIVE full questions, choosing ONE full question from each module.
2. M : Marks , L: Bloom's level , C: Course outcomes.
3. VTU Hand book is permitted.

		Module – 1	M	L	C
Q.1	a.	Explain the construction and working of semiconductor LASER with a neat sketch and energy level diagram.	9	L2	CO1
	b.	Discuss different types of optical fibers based on modes of propagation and RI profile.	6	L2	CO1
	c.	An optical fiber has refractive index of core and cladding of 1.55 and 1.50, respectively. Calculate its numerical aperture and angle of acceptance if it is kept in air.	5	L3	CO5
		OR			
Q.2	a.	Obtain the expression for energy density of radiation in term of Einstein's A and B coefficients.	8	L2	CO1
	b.	Define numerical aperture and derive an expression for numerical aperture of an optical fiber.	7	L2	CO1
	c.	In a diffraction grating experiment the Laser light undergoes first order diffraction at an angle of 19.3°. Find the wavelength of Laser light. Given the grating constant $d = 1.98 \times 10^{-6}$m.	5	L3	CO5
		Module – 2			
Q.3	a.	Setup one dimensional time independent Schrodinger wave equation.	8	L2	CO
	b.	State Heisenberg's uncertainty principle and apply the same to prove the non-existence of free electron inside the nucleus.	7	L2	CO
	c.	An electron is bound in an infinite potential well of width 0.18nm. Find its energy values in the first two allowed energy states.	5	L3	CO2
		OR			
Q.4	a.	Obtain an expression for Eigen function and Eigen energy values for a particle in an infinite potential well of width 'a'.	9	L2	CO2
	b.	What is wave function? Mention the properties of wave function and give its significance.	6	L2	CO2
	c.	Calculate the kinetic energy of a neutron in eV. Given: de –Broglie wave length is 1 Å and mass of neutron, $m_n = 1.674 \times 10^{-27}$Kg.	5	L3	CO2
		Module – 3			
Q.5	a.	Distinguish between classical computing and Quantum computing.	6	L2	CO2
	b.	Explain the CNOT gate and its operation on four different input states.	6	L2	CO2
	c.	Apply Pauli matrices on the state $\|0\rangle$ and $\|1\rangle$.	8	L3	CO2
		OR			
Q.6	a.	Explain the working of T-gate mentioning its matrix representation and truth table.	7	L2	CO2
	b.	Explain Orthogonality and Orthonormality with an example of each.	8	L2	CO2

	c.	A linear operator 'X' operates such that $X\|0\rangle = \|0\rangle$ and $X\|1\rangle = i\|1\rangle$. Find the matrix representation of 'X'.	5		
		Module – 4			
Q.7	a.	Mention the failures of classical free electron theory and explain the assumptions of Quantum free electron theory of metals.	7	L2	CO3
	b.	Explain Meissner's effect and the variation of critical field with temperature.	8	L2	CO3
	c.	A lead wire has a critical field of 6.5×10^5 A/m at 0 Kelvin. The critical temperature is 7.18K. At what temperature the critical field will drop to 4.5×10^5 A/m.	5	L3	CO3
		OR			
Q.8	a.	Define Fermi factor and explain the variation of Fermi factor with temperature and energy.	8	L2	CO3
	b.	Differentiate Type – I and Type – II superconductors.	8	L2	CO3
	c.	Calculate the probability of occupation of an energy level 0.02eV above level at temperature 27°C.	4	L3	CO3
		Module – 5			
Q.9	a.	Explain the importance of (i) size and scale and (ii) weight and strength, in animation.	7	L2	CO4
	b.	Mention the general pattern of Monte – Carlo method and hence explain the procedure to find the value of 'π'.	8	L2	CO4
	c.	In the case of animating a jump, the jump height is 2.5m and jump magnification is 5. Calculate the push height and push acceleration. Given gravitational acceleration is 10m/s.	5	L3	CO5
		OR			
Q.10	a.	Describe jumping and parts of jump.	9	L2	CO4
	b.	Distinguish between descriptive and inferential statics.	6	L2	CO4
	c.	On a particular place, volcanic eruption occurs once in every 100years on an average. Calculate the probability of volcanic eruption in a 100 years interval for K = 0, 1 and 2, assuming the Poisson's model appropriate.	5	L3	CO5

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